Osprey: Pixel Understanding with Visual Instruction Tuning

CVPR'24

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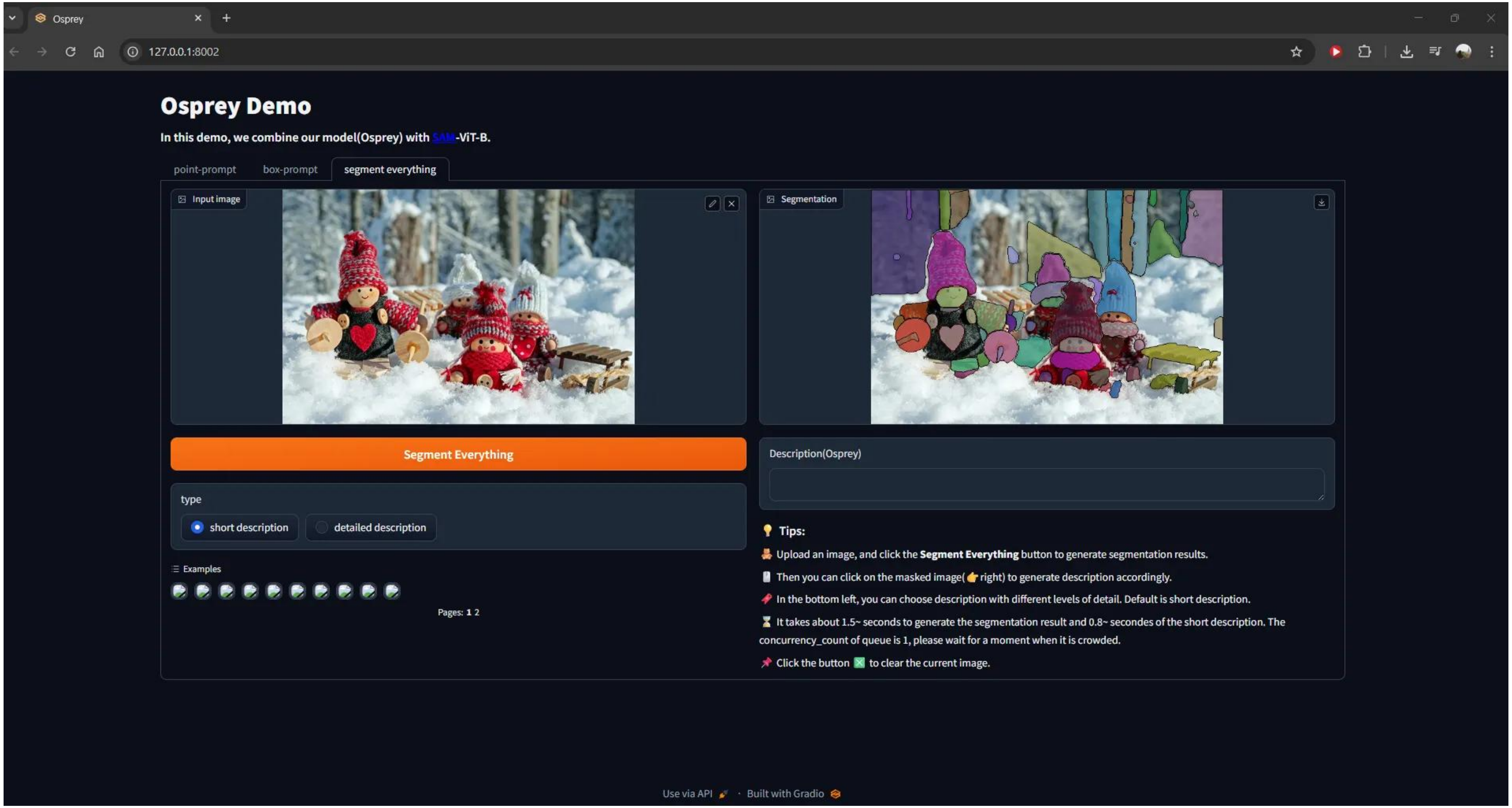
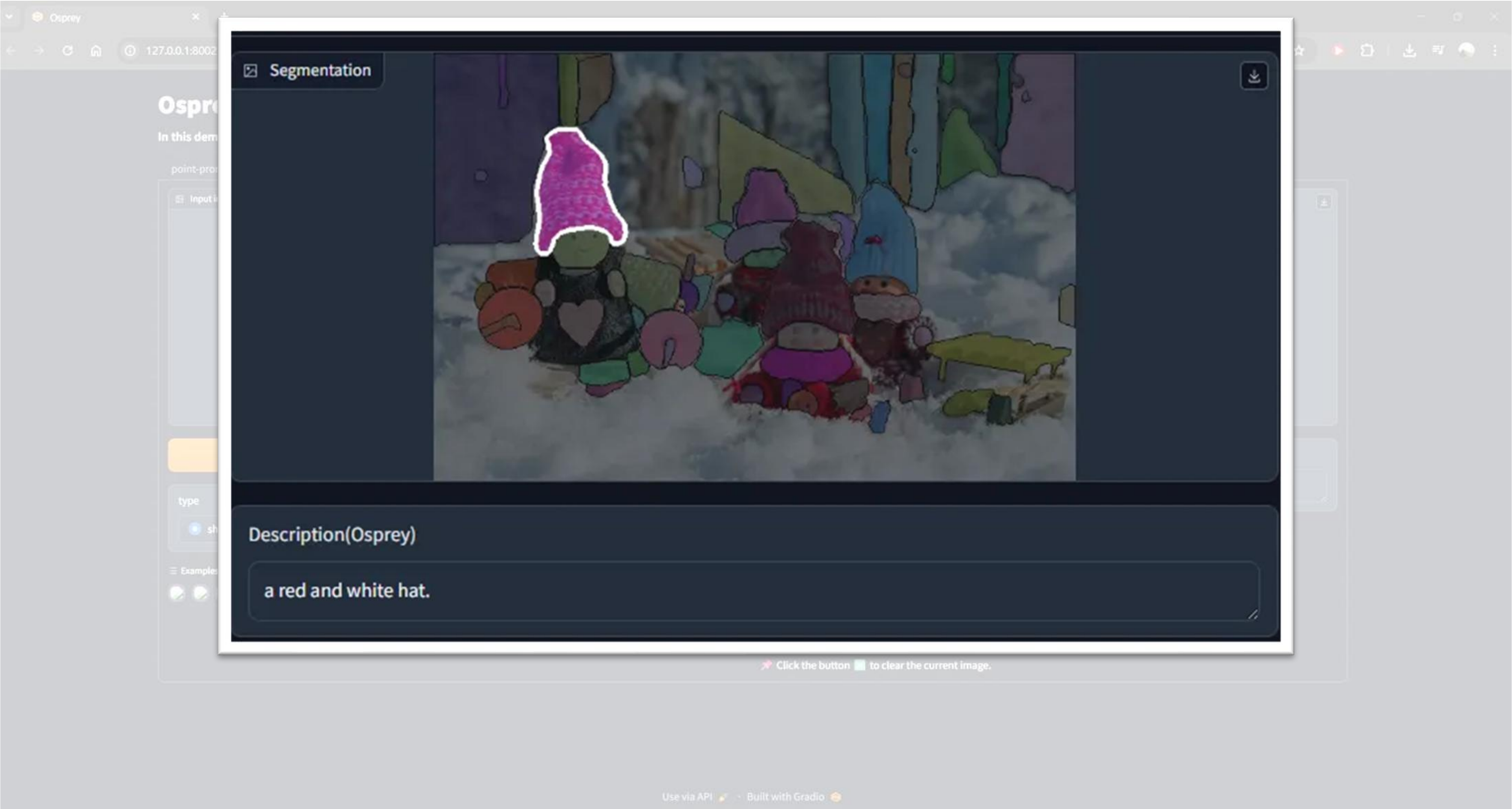
Osprey Demo


Osprey
127.0.0.1:8002
Osprey Demo
In this demo, we combine our model(Osprey) with SAM-ViT-B.
point-prompt
box-prompt
segment everything
Input image
Segmentation
Segment Everything
Description(Osprey)
type
short description
detailed description
Examples
Pages: 1 2
Tips:
Upload an image, and click the Segment Everything button to generate segmentation results.
Then you can click on the masked image(right) to generate description accordingly.
In the bottom left, you can choose description with different levels of detail. Default is short description.
It takes about 1.5~ seconds to generate the segmentation result and 0.8~ secondes of the short description. The concurrency_count of queue is 1, please wait for a moment when it is crowded.
Click the button to clear the current image.
Use via API · Built with Gradio

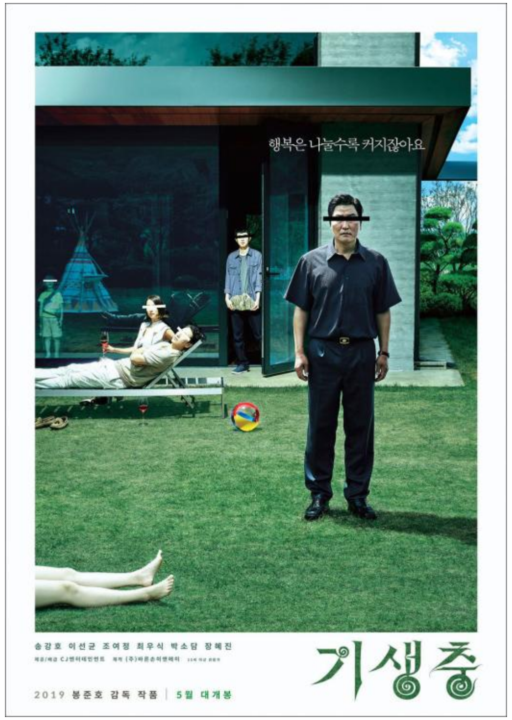
Osprey Demo



Segmentation
Description(Osprey)
a red and white hat.
Click the button to clear the current image.
Use via API · Built with Gradio

MLLM

- General
- Image-level alignment



User:

Describe this image.

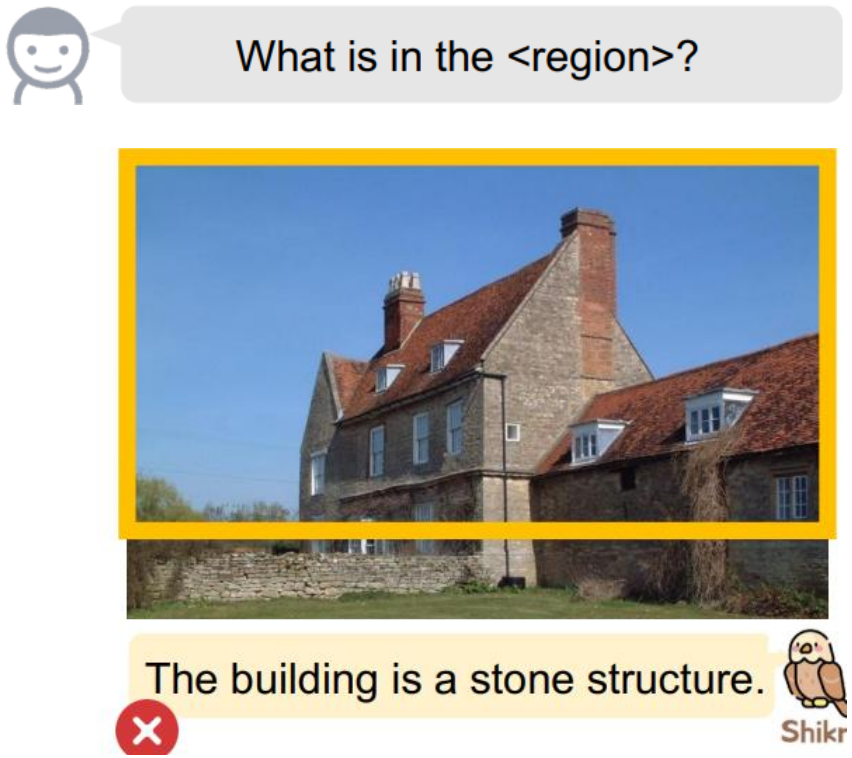
Model Output:

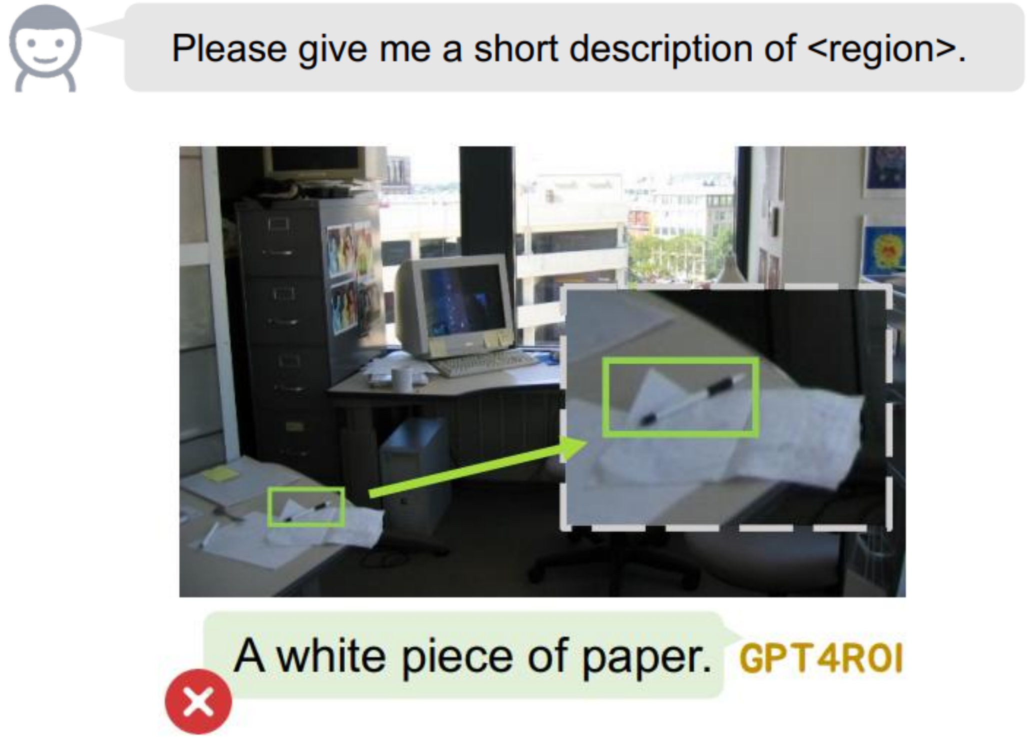
The image features a man standing in a yard, wearing a black shirt and glasses. He appears to be looking at a woman who is lying on a bed in the yard. There are several other people in the scene, some of them standing and others sitting.

In the yard, there is a dining table with a bottle on it, and a sports ball is located nearby. A chair can be seen close to the table, and a handbag is placed on the ground. The scene seems to be a casual gathering or a relaxed outdoor event.

Region-level MLLM

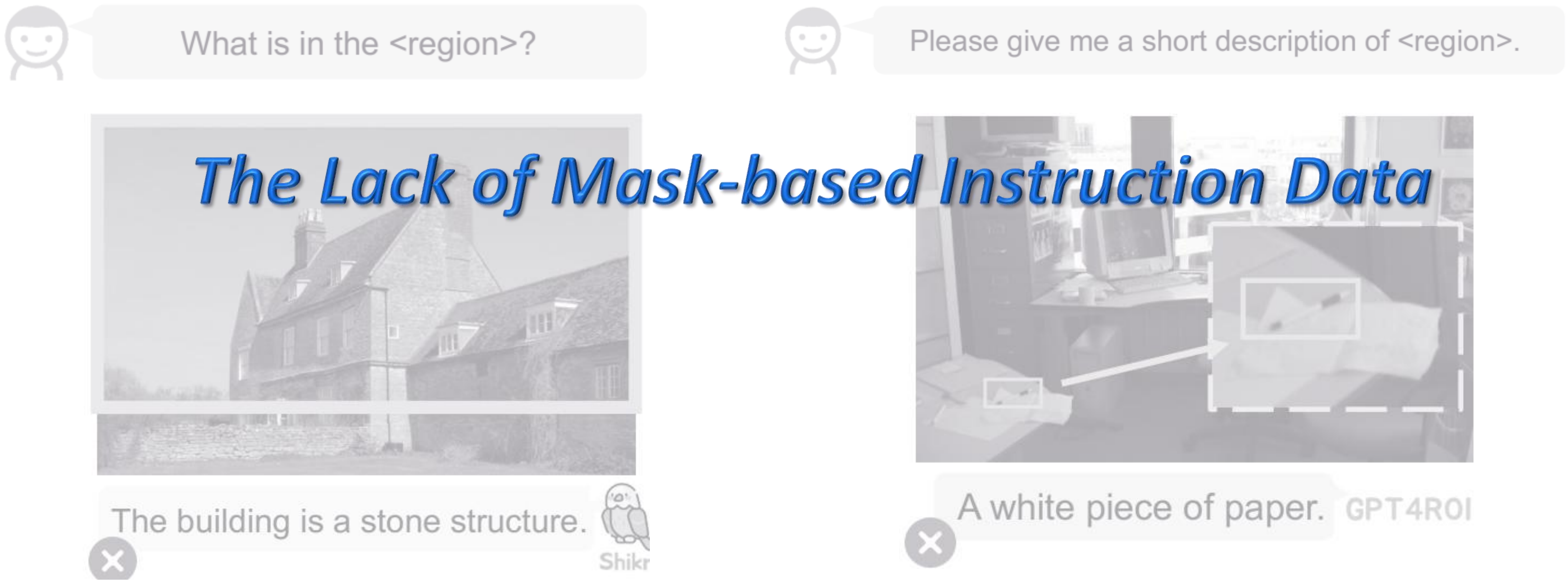
- Kosmos-2, Shikra, PVIT, GPT4RoI
- Box-level, object-level → Sparse(Not fine-grained), Including irrelevant background





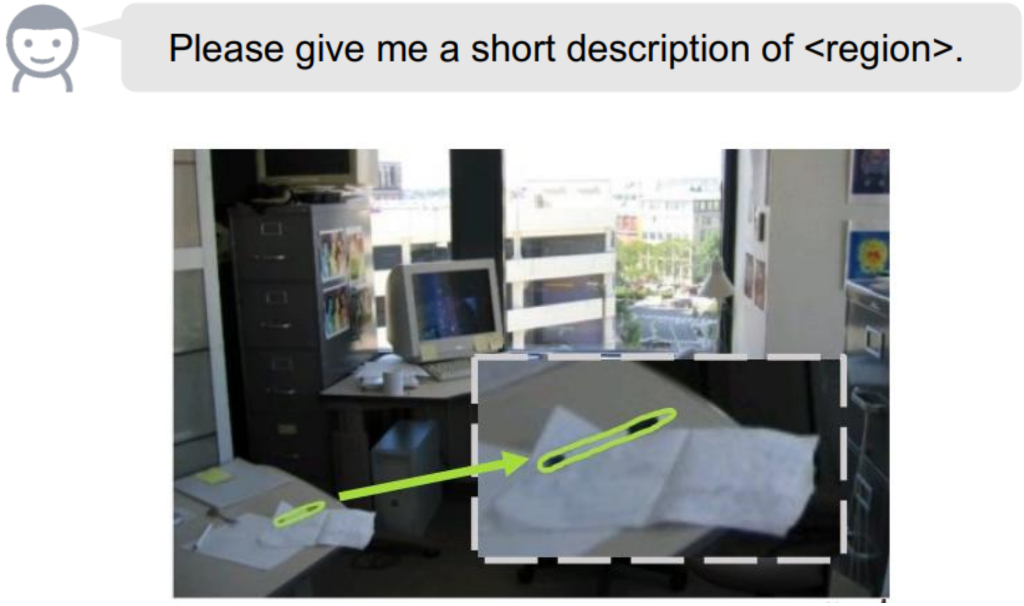
Problem

- Kosmos-2, Shikra, PVIT, GPT4RoI
- Box-level, object-level → Sparse(Not fine-grained), Including irrelevant background



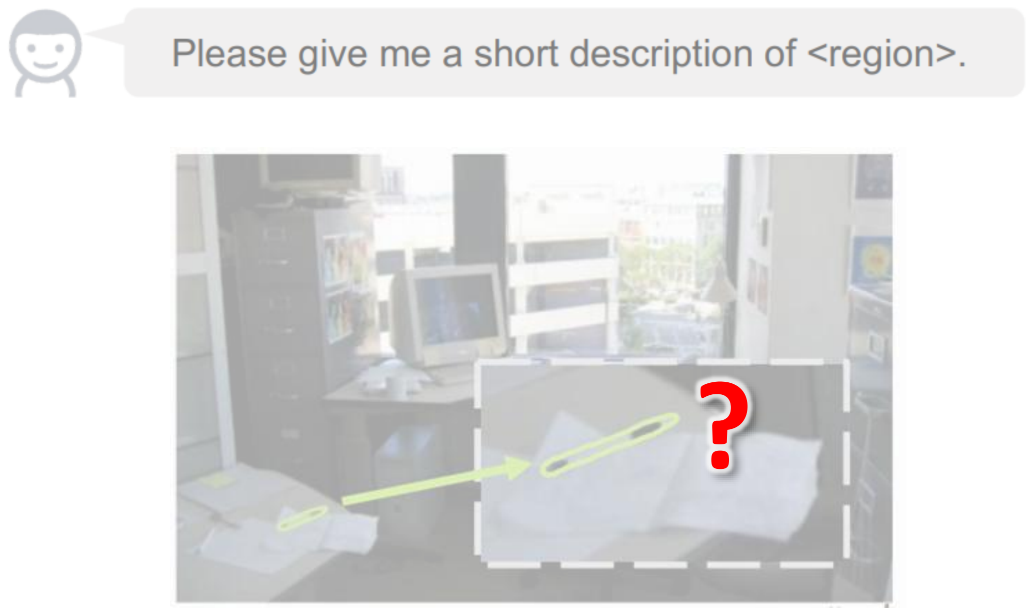
Mask: SAM, HQ-SAM

- Precise objects’ masks, fine-grained segmentation



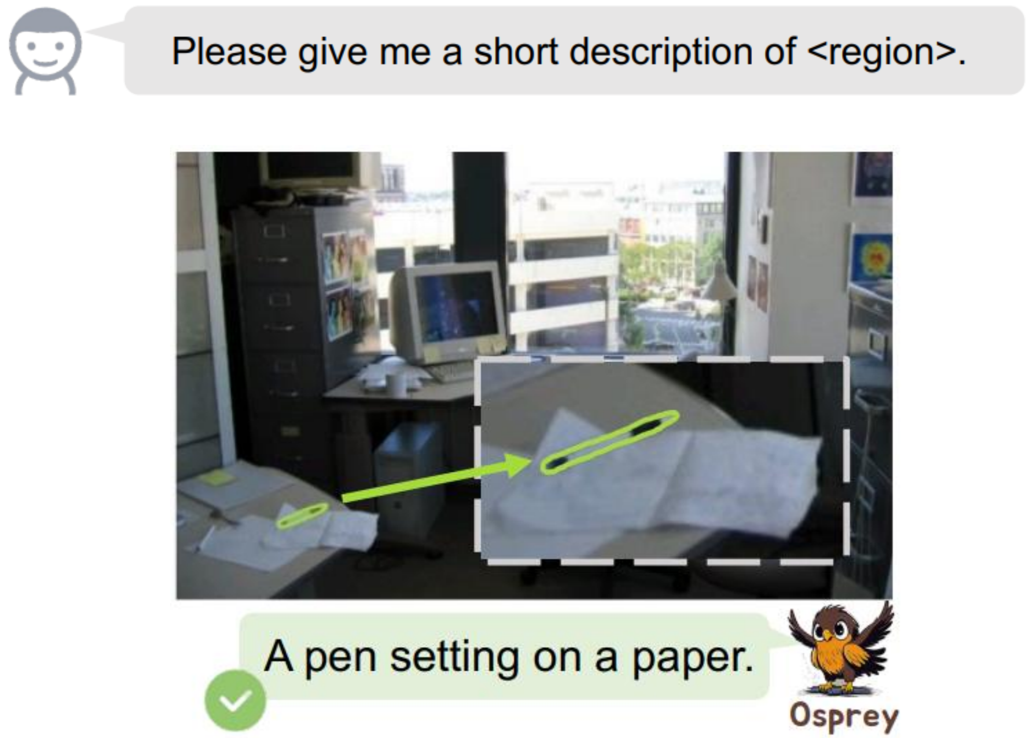
Mask: SAM, HQ-SAM

- Precise objects' masks, fine-grained segmentation
- **NO Semantic labels, detailed attributes and captions** 🙅



Osprey

- A Novel Approach
- Fine-grained Pixel-wise Understanding



Osprey-724K

- To obtain fine-grained pixel-level alignment between vision & language features
- Curate a large scale mask-based region-text dataset

Osprey-724K: Object-level Instructions

- Public Dataset; COCO, RefCOCO, RefCOCO+, RefCOCOg

→ Plain & Short, few semantic context ⇒ **Insufficient to train an MLLM!**

Osprey-724K: Object-level Instructions

- Public Dataset; COCO, RefCOCO, RefCOCO+, RefCOCOg

  → Plain & Short, few semantic context ⇒ **Insufficient to train an MLLM!**

Generate fine-grained region-based instruction data

Object category, object type, object action, location, color, status etc.

Osprey-724K: Object-level Instructions

Input for GPT-4

- Global Context: LLaVA-115K(COCO)
- Local Hint: Bbox, brief region caption(RefCOCO,+,g)

GPT-4 Prompting and Generation

- Prompt: “You are an AI visual assistant that can analyze a single image. ~~~”
- Generate 2 types of data (197K)
  - Region level Detailed Description
  - Conversation Examples (Q&A)

Osprey-724K: Part-level Instructions

- To capture the part-level knowledge
- PACO-LVIS

Object categories	Object-specific part classes	Attribute	Color	Pattern & markings	Materials	Reflectance
75	456	55	29	10	13	3

- GPT-4 Generates QA data (306K)

Question	What is in <region1>?
Answer	Spoon
Question	What is the color of the <region2>?
Answer	Dark green

Robustness and Flexibility

Negative Sample Mining to mitigate hallucination

- LVIS (1,200 object categories)
- Pos/Neg samples equally balanced
- "Yes/No" response strategy
- Hard Negative Sampling; No random, Confusable samples
  - Spatial-aware: the nearest object
  - Class-aware: semantically similar
    e.g. "The category of <region> is a dog, right?" – "No"

Robustness and Flexibility

Short-form Response Instructions for flexibility

- Add short-form response instructions
  - Using the same dataset, e.g. COCO etc.
  - Categories, colors, types, locations or quantities of a specific object region
- GPT-4 is required to produce a single word or phrase
- When a short-form response is needed, use an explicit prompt
  (e.g. “Answer in single word.”)

Data statistics of the Osprey-724K

Type	Form	Raw Data	GPT-4	#Samples
Object-level	Descriptions Conversations	COCO/RefCOCO/RefCOCO+/ RefCOCOg/LLaVA-115K	✓ ✓	70K 127K
Part-level	Categories Attributes	PACO-LVIS	✓ ✓	99K 207K
Robustness &Flexibility	Positive/Negative Short-Form	COCO/RefCOCO/RefCOCO+/ RefCOCOg/LLaVA-115K/LVIS	✗ ✓	64K/64K 99k

730K...?

Table 1. Data statistics of Osprey-724K.

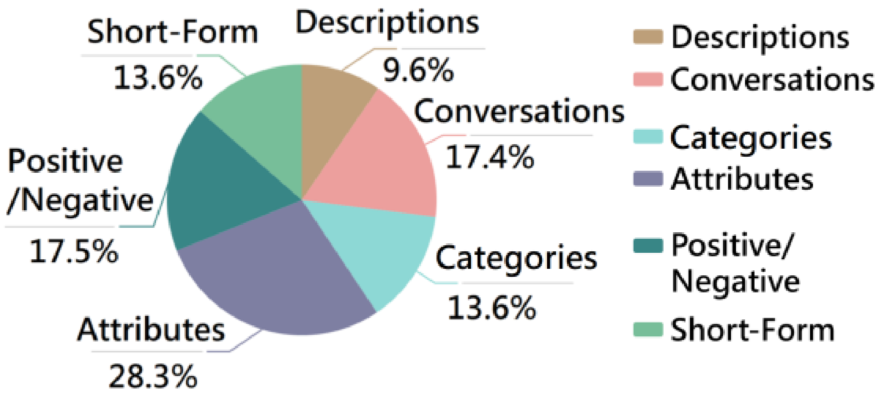


Figure 3. Data distribution of Osprey-724K.

Example sample of the Osprey-724K

Object Categories

1. dog
2. dog collar
3. tag
4. chair
5. cushion
6. dinner table
7. cookie
8. cookie
9. cake
10. box

Part Attributes

11 Q: What is the category?
A: Side of a box.
Q: What is the material?
A: Paper.

12 Q: Is the inner side of the box plain in <region12> ?
A: Yes.
Q: What is the color?
A: White.

Detailed Descriptions

4 The chair behind the dog has a brown color and appears to be made of wood. It has a simple design that complements the cushions' blue color on which the dog is resting.

9 In front of the dog and on the dining table, there are several decorative dog treats and cakes. One of them looks like a big white bone, while others have dots on them. There is also a dotted item in front of the treats on the left side.

...

Robustness and Flexibility

1 Q: Cat is the category of <region1>, right?
A: No, there is no cat in the region.

2 Q: Is the category of <region2> dog?
A: No, it isn't.

4 Q: Where is the wood part of the chair located in <region4>? Answer the question using a single word or phrase.
A: Behind the dog.

Conversations and Complex Reasoning

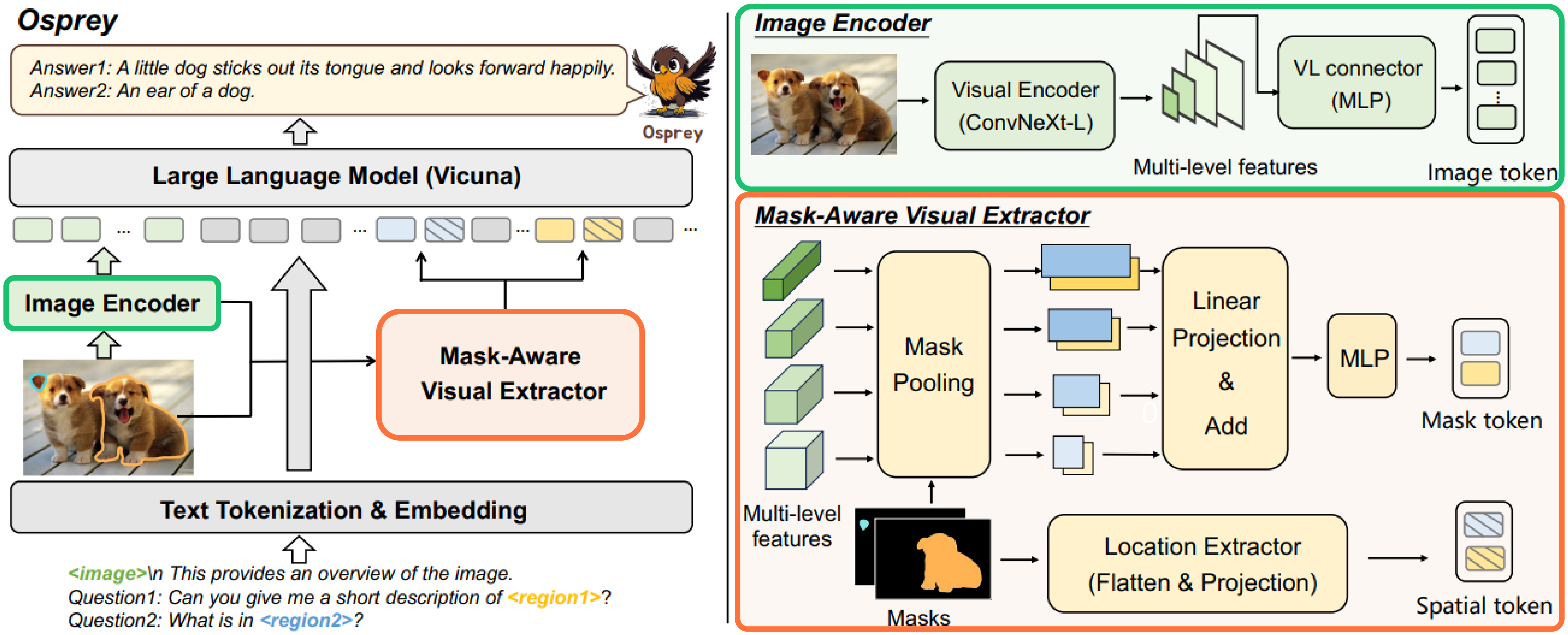
Question: Can you describe the pillow that the dog is resting on in <region5>?
Answer: The pillow in <region5> is a blue cushion or pillow that is part of the chair. It provides a comfortable resting place for the dog. The cushion is visible in the bottom part of the image and supports the dog's brown paws as it rests on the chair.

Question: What can you tell me about the items on the cake in <region7>, <region8> and <region9>?
Answer: In all regions, there are treats or cakes displayed. In <region9>, there is a plated item with dots or a pattern on it, placed above a big white bone. It seems to be a celebratory arrangement of doggy treats and cakes. Meanwhile, in <region7> and <region8>, there are two small dog treats or bones sitting on a plate. These treats may be specially made for dogs with some writing or decoration on them.

Question: Can you provide any more details about the dog and its surroundings based on the descriptions?
Answer: The dog in the image is black and brown and is sitting on a blue chair cushion, as seen in <region5>. The dog is surrounded by celebratory doggy treats and cakes, as mentioned in <region7> and <region8>. The presence of the treats and the dog's position on the chair suggests a special occasion or celebration. The overall theme of the image seems to be centered around the dog and its enjoyment of the treats and cakes.

Figure 2. Example sample of the Osprey-724K dataset to illustrate the mask-based instruction-following data.

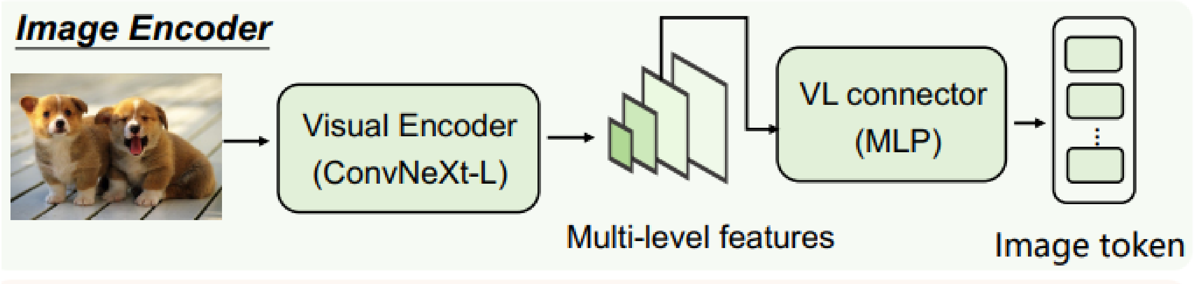
Overview of Osprey



Convolutional CLIP Vision Encoder

Generally, Pixel-level tasks work better in high resolutions

- Image resolution of ViT CLIP: 224×224 or 336×336 → TOO LOW
  - Resolution↑ → Computational cost ⇑ in ViT



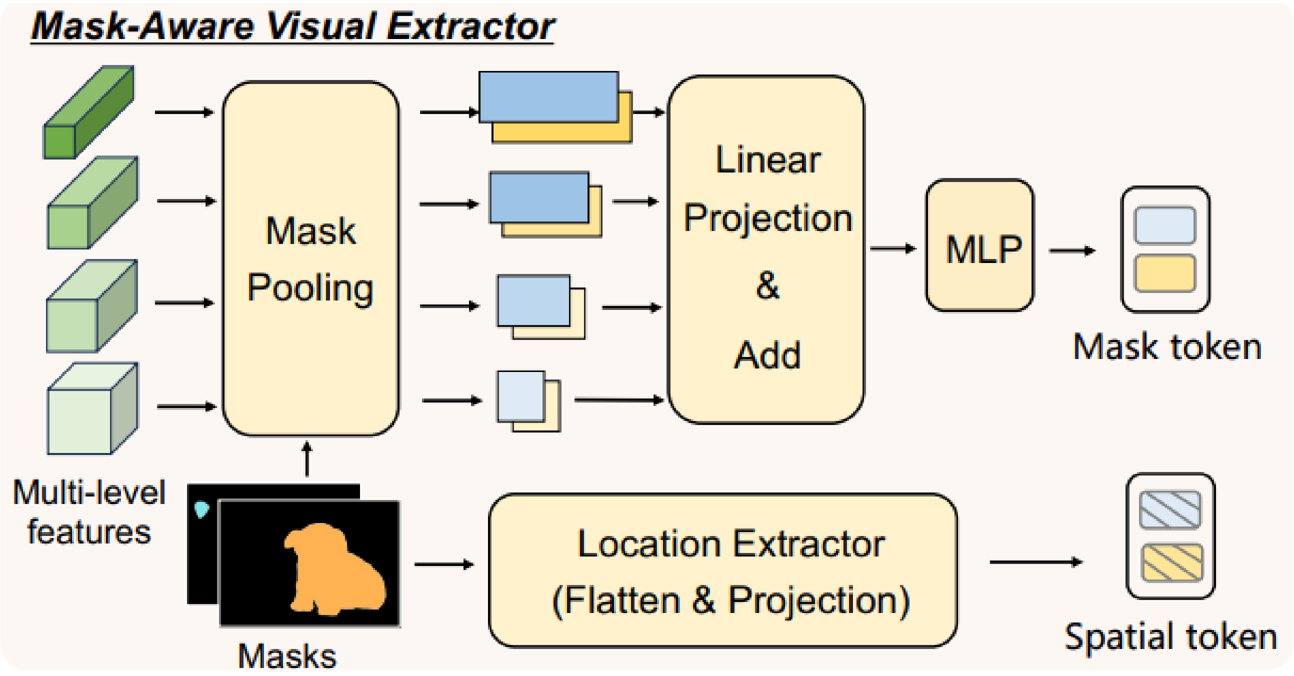
- Introduce ConvNeXt backbone
  - Robust to various input resolution, Efficient at high resolution
  - Multi-scale Feature Map
  - Image features are extracted from ConvNext-L "res4" stage

Mask-Aware Visual Extractor

Motivation: Fine-grained Representation

Captures both **visual semantic features** and **spatial geometry** of the target region.

- Visual Mask Token(t_i)
- Spatial Token (s_i)



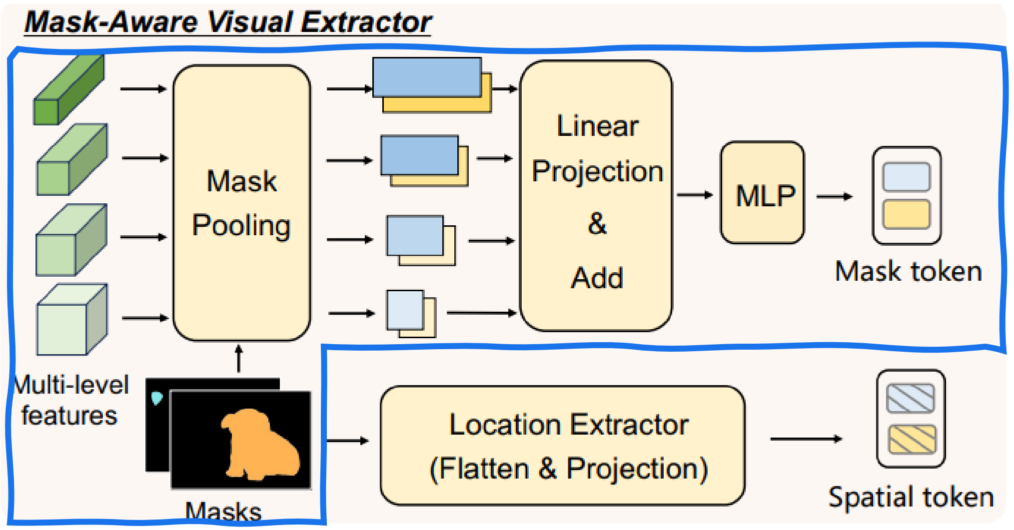
Mask-Aware Visual Extractor

Visual Mask Token(t_i)

- Source: Multi-level features from ConvNeXt encoder $\mathbf{Z}(x)_j$
- Process:
  - Mask Pooling: Extract features only within the mask region ($\mathbf{R}_i$)
  - Fusion: Linear Projection ($\mathbf{P}_j$) → Summation → MLP (σ)
- Output: Captures semantic details across scales.

$$V_{ij} = \mathcal{MP}(\mathbf{R}_i, \mathbf{Z}(x)_j).$$

$$t_i = \sigma(\sum_{j=1}^{4} \mathbf{P}_j(V_{ij})).$$

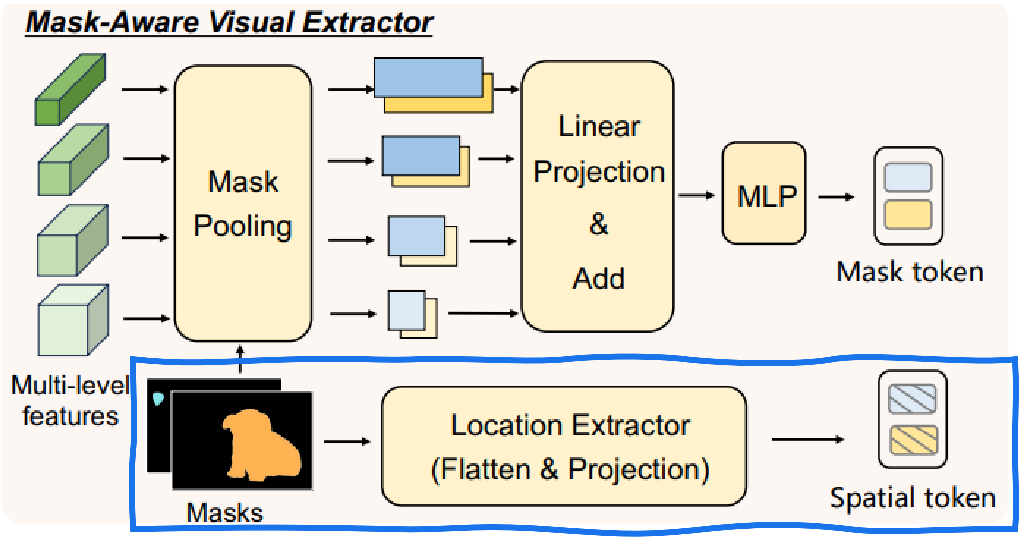


Mask-Aware Visual Extractor

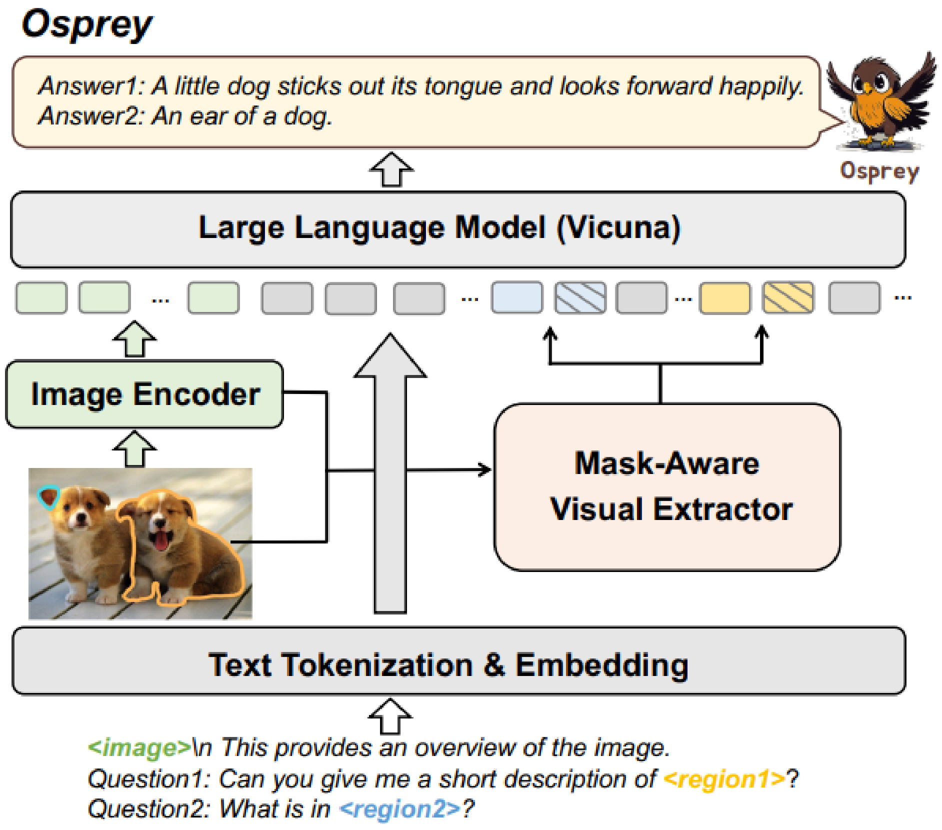
Spatial Token (s_i)

- Source: Binary Mask $$\mathbf{M}^{H \times W} \in \{0, 1\}$$
- Process: Resize to 224×224 → Flatten → Linear Projection
- Output: Encodes precise spatial geometry relationships

Region Embedding: Visual Mask Token(t_i) + Spatial Token (s_i)



Tokenization for LLM Model



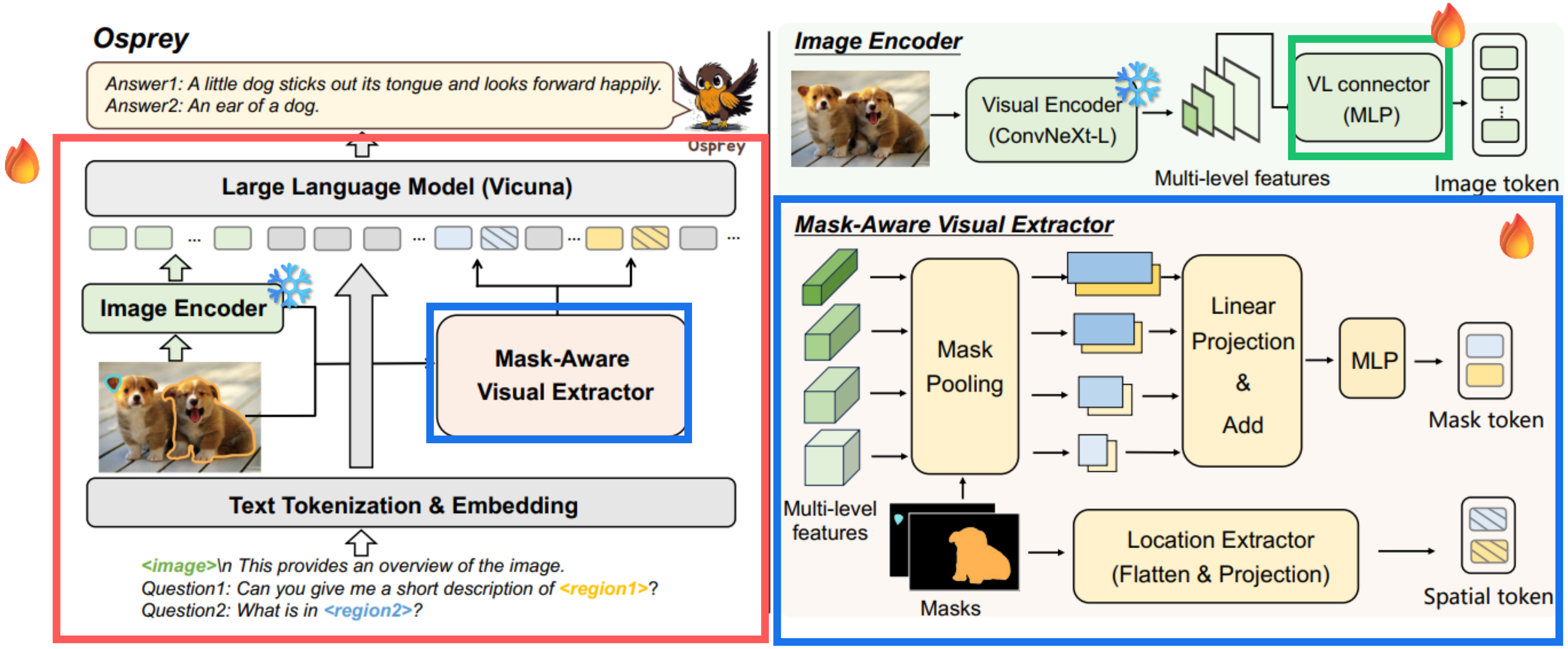
- Image Token: image-level embed. (ConvNeXt “res4”)
- Text Token: User query, “Describe the <region>”
- Region Token:
  <region> → replaced Visual token + Spatial token
- Prefix prompt:
  “<image>\n This provides an overview of the picture.”

<image>\n This provides an overview of the image.
Question1: Can you give me a short description of <region1>?
Question2: What is in <region2>?

- LLM: Vicuna (LLaMA)

Training

1. Image–Text Alignment Pre-training: 🔥 Image-level projector ❄️ Vis Encoder, LLM
   CC3M
2. Mask–Text Alignment Pre-training: 🔥 Mask-Aware Visual Extractor
   short text and Object-level(COCO, RefCOCO, RefCOCO+)/Part-level(Pascal Part, Part Imagenet)
3. End-to-End Fine-tuning: 🔥 Image projector + Mask Extractor + LLM ❄️ Vis Encoder
   Osprey-724K/VG(Visual Genome)/VCR(Visual Commonsense Reasoning)

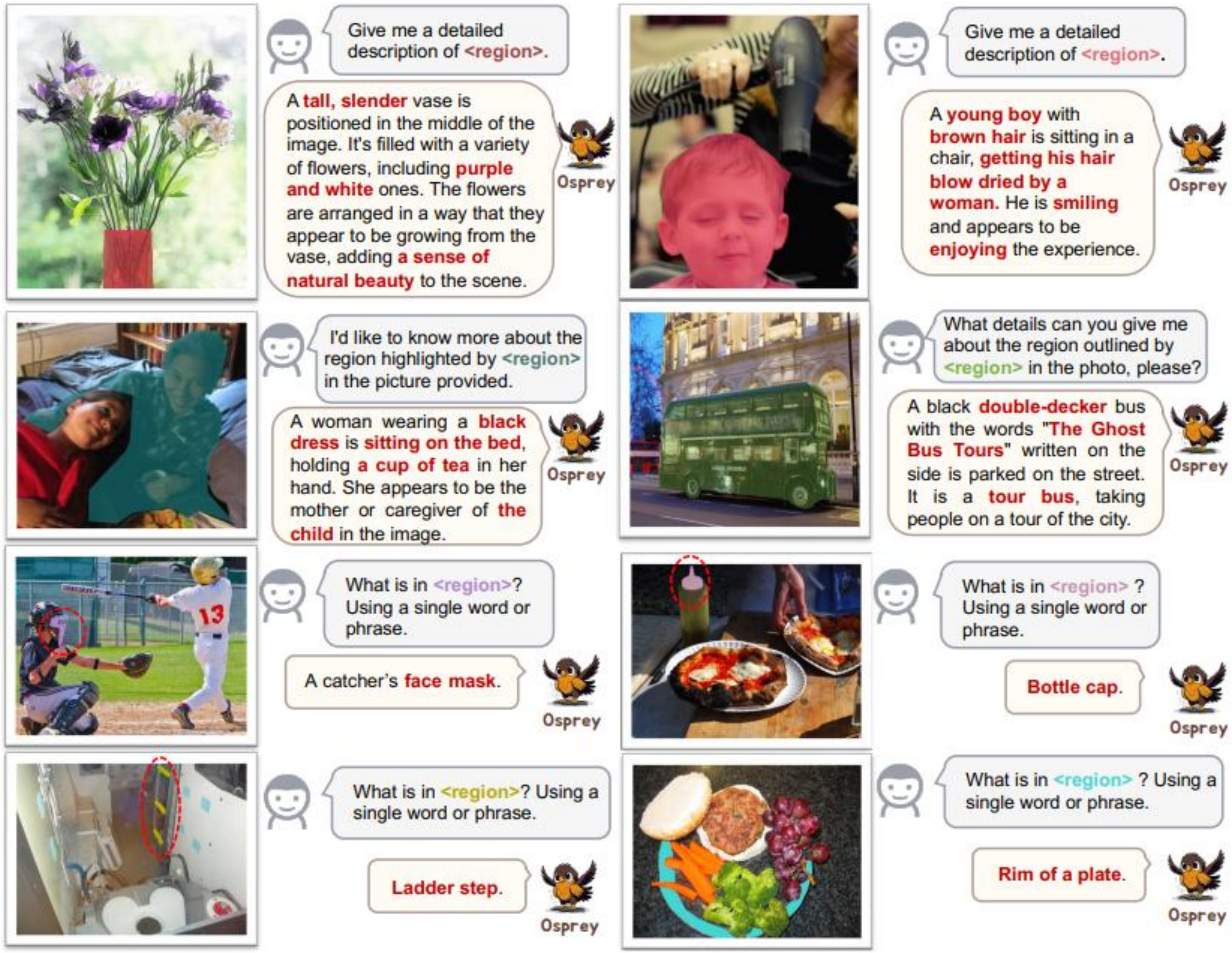


Experimental setup

- Input Resolution: 512×512
- Max Sequence Length: 2,048
- 4 NVIDIA A100 (80GB)
- Training time: ~70h
- Optimizer: AdamW
- Scheduler: Cosine Annealing

Stage	Batch size	LR	Epoch
1. Image-Text	128	1×10^{-3}	1
2. Mask-Text	4	2×10^{-5}	1
3. End-to-End	4	1×10^{-5}	2

Visual results on the input mask-based regions



Visual results with SAM

1. The dirt road is visible in the image, and it appears to be a long and winding one. The road is surrounded by tall grass and bushes, and there are tire tracks visible on the dirt.
2. The sky is clear and blue, with no clouds visible. The sun is shining brightly, and the sky is free of clouds.
3. The right side of the image shows a tall grass field, with the tallest grass located on the far right side of the road. The grass is dry and straw-like, and the road appears to be cut off at the edge of the field.
4. A white van is observed driving on a dirt road, situated in the middle of a grassy field. The vehicle appears to be in motion, traveling through the countryside.
5. The tail light of a car is visible in the image, indicating that the vehicle is in motion. The car is a white van, and the tail light is on.

1. A Statue of Liberty is visible in the image, located in front of a building. The statue is depicting a famous landmark in New York, with a lit-up torch on top. The statue is made of stone and appears to be a popular tourist attraction.
2. The white base of the statue of liberty is visible in the image. It appears to be a large concrete base, possibly used to support the statue.
3. The building in the background has several windows, and the leftmost window is visible in the image. It appears to be a rectangular window with a white frame.
4. A green tree can be seen in the background of the image, located to the right of the radio tower. It is the closest tree to the tower and is situated in the middle of the trees.

Open-vocabulary categories | Short descriptions | Detailed descriptions

Open-Vocabulary Segmentation

“Can you give me a short description of <region>? Using a short phrase.”

Ground-truth mask regions를 mask 입력으로 사용, 모델이 해당 영역을 **어떻게 인식하는지**만을 평가
모델의 output은 문장 형태이므로 Sentence-BERT를 이용하여 Semantic Similarity 계산, 가장 유사도가 높은 카테고리 채택

Method	Type	Cityscapes			ADE20K-150		
		PQ	AP	mIoU	PQ	AP	mIoU
CLIP-ConvNeXt-L [43]	Mask	22.53	12.07	23.06	36.86	39.38	28.74
CLIP-Surgery-ViT-L [30]	Mask	27.24	28.35	21.92	26.55	29.70	21.42
Kosmos-2 [40]	Box	12.09	9.81	13.71	6.53	4.33	5.40
Shikra-7B [5]	Box	17.80	11.53	17.77	27.52	20.35	18.24
GPT4RoI-7B [58]	Box	34.70	21.93	36.73	36.32	26.08	25.82
Ferret-7B [54]	Mask	35.57	26.94	38.40	39.46	29.93	**31.77**
Osprey-7B (Ours)	Mask	**50.64**	**29.17**	**49.78**	**41.89**	**41.24**	29.63

Table 2. Recognition performance on open-vocabulary panoptic segmentation (PQ), instance segmentation (AP) and semantic segmentation (mIoU) upon the validation sets of Cityscapes [11] and ADE20K [59]. The ground truth box/mask is used for performance evaluation.

Referring Object Classification

이미지 내 특정 영역(Specific Region)에 있는 객체가 무엇인지 분류

"What is the category of <region>? Using only one word or phrase."

SS(Semantic Semilarity): 예측한 레이블과 GT레이블의 Semantic Space 상에서 얼마나 유사한지 측정. e.g. '강아지', '개'는 유사도가 높음
S-IoU(Semantic IoU): 예측한 단어들과 정답 단어들 간의 Overlap 정도를 반영

Method	LVIS		PACO	
	SS	S-IoU	SS	S-IoU
LLaVA-1.5 [32]	48.95	19.81	42.20	14.56
Kosmos-2 [40]	38.95	8.67	32.09	4.79
Shikra-7B [5]	49.65	19.82	43.64	11.42
GPT4RoI-7B [58]	51.32	11.99	48.04	12.08
Ferret-7B [54]	63.78	36.57	58.68	25.96
Osprey-7B (Ours)	**65.24**	**38.19**	**73.06**	**52.72**

Table 3. Semantic similarity and IoU results of referring object classification on *object-level* LVIS and *part-level* PACO. SS/S-IoU denotes Semantic Similarity/IoU, respectively.

Referring Description & Reasoning

Detailed Description: 모델이 User의 지시에 따라 특정 영역을 얼마나 상세하게 묘사할 수 있는지 평가, GPT-4 Judge
Ferret-Bench: 단순 묘사를 넘어 복잡한 추론 능력을 평가하기 위함, mask annotation 없음 → Box로 대체

데이터 생성 파이프라인을 사용하여 평가용 질문과 GPT-4로 모범 답안을 생성

Method	Detailed Description
LLaVA-1.5 [32]	71.11
Kosmos-2 [40]	40.89
Shikra-7B [5]	40.97
GPT4RoI-7B [58]	49.97
Osprey-7B (Ours)	77.54
Osprey-7B* (Ours)	**83.78**

Table 4. Detailed region description performance evaluated by GPT4 on the validation set of RefCOCOs. ∗ denotes the model trained with additional part of data mixture of 665K samples used in LLaVA-1.5 [32] in Stage 3 (the same below).

Ferret-Bench	Osprey-7B*	Ferret-7B	Kosmos-2	Shikra-7B
Referring Description	**72.2**	68.7	51.8	46.0
Referring Reasoning	**67.8**	67.3	33.7	41.6

Table 5. Results on Ferret-Bench [54]. Note that *we use box as the input region* due to lack of mask annotations on Ferret-Bench.

Object Hallucination

Random: 이미지에 없는 객체를 무작위로 물어보는 설정
Popular: 데이터셋에 자주 등장하는 객체가 있는지 물어보는 설정 (Bias test)
Adversarial: 모델이 헷갈릴 만한 동시 등장 확률이 높은 객체를 물어보는 적대적 설정

Sampling	**Metrics**	Osprey-7B*	Ferret-7B	Shikra-7B	LLaVA-1.5	InstructBLIP	MiniGPT4	MM-GPT	mPLUG-Owl
Random	Accuracy	89.47	**90.24**	86.90	88.73	88.57	79.67	50.10	53.97
	Precision	93.40	97.72	94.40	88.89	84.09	78.24	50.05	52.07
	Recall	84.93	83.00	79.26	88.53	95.13	82.20	100.00	99.60
	F1 Score	88.97	89.76	86.19	88.71	89.27	80.17	66.71	68.39
	Yes (%)	45.47	43.78	43.26	49.80	56.57	52.53	99.90	95.63
Popular	Accuracy	**87.83**	84.90	83.97	85.83	82.77	69.73	50.00	50.90
	Precision	89.94	88.24	87.55	83.91	76.27	65.86	50.00	50.46
	Recall	85.20	80.53	79.20	88.67	95.13	81.93	100.00	99.40
	F1 Score	87.50	84.21	83.16	86.22	84.66	73.02	66.67	66.94
	Yes (%)	47.37	45.63	45.23	52.83	62.37	62.20	100.00	98.57
Adversarial	Accuracy	**85.33**	82.36	83.10	72.10	65.17	79.20	50.00	50.67
	Precision	85.43	83.60	85.60	74.69	65.13	61.19	50.00	50.34
	Recall	85.20	80.53	79.60	88.34	95.13	82.93	100.00	99.33
	F1 Score	85.31	82.00	82.49	80.94	77.32	70.42	66.67	66.82
	Yes (%)	49.87	48.18	46.50	59.14	73.03	67.77	100.00	98.67

Table 6. Results on the object hallucination benchmark across three evaluation settings of POPE [29] benchmark.

Region Level Captioning

“Please give me a short description of <region>.”

Osprey가 특정 영역에 대해 얼마나 **자연스럽고 정확한 문장(Caption)을 생성**할 수 있는지를 평가

METEOR, CIDEr: Caption 생성 품질 평가 점수

Method	Type	METEOR	CIDEr
GRIT [50]	Box	15.2	71.6
Kosmos-2 [40]	Box	14.1	62.3
GLaMM [45]	Box	16.2	105.0
Osprey-7B (Ours)	Mask	**16.6**	**108.3**

Table 7. Region captioning performance evaluated on the validation set of RefCOCOg.

Ablation Study

Different Vision Encoders

Method	Cityscapes	ADE	LVIS		PACO	
	PQ	PQ	SS	S-IoU	SS	S-IoU
ViT-L	38.58	38.86	60.89	31.02	70.23	48.57
ConvNeXt-B	48.49	41.94	64.52	37.02	72.86	51.62
ConvNeXt-L	**50.64**	**42.50**	**65.24**	**38.19**	**73.06**	**52.72**

Table 8. Comparisons with various vision encoders on Open-Vocabulary Segmentation and Referring Object Classification.

Various Input Image Sizes

Input	#Image Tokens	Speed	SS	S-IoU
224	196	**6.0**	53.20	26.12
336	441	5.8	56.70	28.90
512	1024	3.5	65.24	38.19
800	2500	1.9	**68.29**	**42.66**

Table 9. Comparisons across various input image sizes of ConvNeXt-based CLIP vision encoder on LVIS [15]. Note that *the speed is measured by the number of input mask-text pairs processed per second* during model inference. The evaluation is conducted on a single NVIDIA A100 GPU.

Impact of Short-form prompt & Pos/Neg Data

Method	**LVIS**		**PACO**	
	SS	S-IoU	SS	S-IoU
w/o Short-form	56.41	25.65	50.26	23.29
w/o Pos./Neg.	63.55	36.70	71.59	50.39
Osprey-724K	**65.24**	**38.19**	**73.06**	**52.72**

Table A4. Performance comparisons with and without short-from prompt and positive/negative samples on *object-level* LVIS [15] and *part-level* PACO [44].

Single-level vs. Multi-level mask features

Method	Cityscapes	ADE	LVIS		PACO	
	PQ	PQ	SS	S-IoU	SS	S-IoU
Single-level	46.28	38.03	62.46	34.25	68.42	46.38
Multi-level	**50.64**	**42.50**	**65.24**	**38.19**	**73.06**	**52.72**

Table A2. Comparison between single-level and multi-level mask features in Osprey model training.

Conclusion

- Osprey: Pixel-level mask region references + language instruction

  → Enhanced MLLMs for fine-grained visual understanding
- Curated the Osprey-724K; 724K mask-based region-text pairs
- SOTA